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BENGALURU CAMPUS

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Intent-Based SDN for 5G SLA Enforcement:

Final Project Report

TEAM – 16

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Abstract
We implement and evaluate a transport-network service-assurance prototype. An operator submits a structured intent, the system validates it and checks residual resources, computes a feasible route, programs software switches, measures the service, and responds to confirmed violations. Strict intents use versioned consistent updates; relaxed intents use direct replacement. The results show the benefit of preserving packets during tested route transitions and suppressing oscillation, together with costs in reaction time, rule space, reordering, and control-stack CPU. The artifact integrates established methods and makes no algorithmic novelty claim. Demonstrations run on a live Mininet/Open vSwitch datapath inside Docker; illustrative dashboard samples are explicitly labelled and excluded from evaluation.

1 Introduction and objectives
Manual switch configuration requires an operator to map a service request onto routes, matches, ports, queues, and subsequent repairs. A service promise also needs ongoing measurement: installing a rule does not prove the requested service is delivered. This project makes that lifecycle explicit and reproducible on one machine.

Our objectives are to validate structured requests with explainable refusals; account for existing reservations; choose constraint-satisfying routes; enforce matches and bandwidth queues; detect sustained service violations; replan through a durable event boundary; and expose an understandable audit trail. Evaluation measures both benefits and costs of consistency and damping. The scope is a transport-network reference implementation motivated by 5G-style services, rather than a deployed radio or mobile-core system.

2 Literature review
Leivadeas and Falkner [1] survey the intent lifecycle, including expression, translation, resolution, activation, and assurance. Their discussion motivates a feedback loop that measures service behavior after activation. We instantiate those stages using a frozen structured grammar and path-level remediation. Natural-language interpretation and machine-learning assurance are outside the implementation.

Martini, Gharbaoui, and Castoldi [2] describe intent-based vertical-service slicing with feasibility checks, translation, deployment, and assurance over cloud and NFV infrastructure. Their recovery redeploys a slice. Our corrective action changes a transport path, so the workloads and timing scopes differ; we do not claim a direct speedup over their system.

Park and colleagues [3] review SDN and assurance across network-slicing domains. Their hybrid monitoring discussion motivates active observations together with passive counters. Our planner uses bottleneck bandwidth and summed link delays inside one transport domain. The implementation does not include RAN scheduling, a mobile core, FlexE, or segment routing.

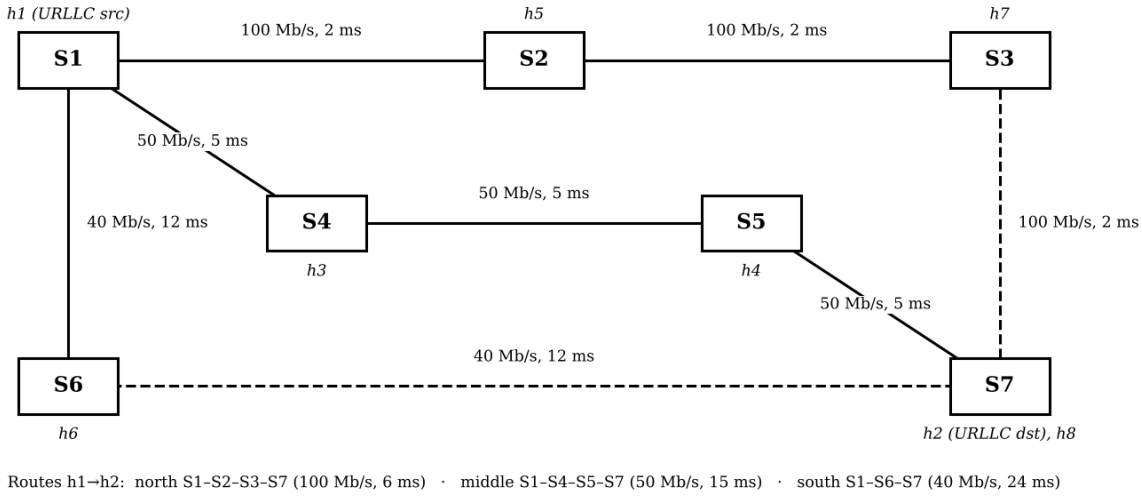
Foerster, Schmid, and Vissicchio [4] survey consistent network updates and the tradeoffs among correctness properties, update rounds, and extra rules. This provides the design space for our strict and relaxed modes. A software-switch rule-capacity pilot cannot substitute for hardware TCAM evaluation.

Reitblatt and colleagues [5] establish consistent-update abstractions and a versioned two-phase method. Our strict implementation stages tagged rules, synchronizes processing, flips ingress classification, drains the old path, and removes the old version. The method is cited as prior work. The emulated drain interval is a deployment assumption rather than a general proof about packet lifetime.

Zhou and colleagues' CCG [6] enforces customizable consistency properties with verification and update scheduling. Our per-intent strict/relaxed choice follows the principle that stronger consistency has a cost. We do not implement CCG's invariant verifier, uncertainty model, or automatic fallback. These distinctions prevent presenting an educational integration as a new consistency algorithm.

3 System design
The processing pipeline is YAML or JSON intent -> M1 validation and persistence -> M2 admission -> M3 constrained path -> M4 rules and queues -> live software datapath -> M5 measurements and detector -> durable violation -> control-loop replanning. M6 reads the shared state and causal audit. SQLite provides the process boundary between telemetry and the Ryu controller. These are logical modules, not a requirement for a separate server per module. Compose provides the privileged SDN runtime, an API service, a read-only dashboard service, and a one-shot seed job. The Ryu controller and live collector run inside the SDN runtime during scenarios.

The topology is defined in topology/topology_spec.py and instantiated by topology/team16_topo.py. It contains seven switches, eight hosts, eight inter-switch links, and eight host-access links. A single Ryu controller programs the OVS switches. Three link-disjoint alternatives between the protected hosts have unequal budgets so the system has a meaningful selection problem. The s3-s7 and s6-s7 links are marked inferred and remain modelling assumptions. These counts describe configuration, rather than measured performance.



Topology reproduced from the supplied Team 16 literature report with inferred links dashed

The frozen grammar contains traffic endpoints and protocol, optional destination port, priority, service guarantees, consistency, violation policy, and dwell. Supported guarantee types are minimum bandwidth, maximum latency, maximum loss, isolation from selected tenants, and avoided links. Actions are reroute, degrade, or alert_only. The schema and semantic validator define acceptance; no LLM interprets the request. Application names such as remote surgery label illustrative configurations and do not certify a clinical or standards-compliant service class.

3.1 Standards-derived service profiles

This front-end layer generates ordinary intents and then invokes the unchanged validation/admission handler. The schema and topology remain unchanged. A 5QI describes QoS characteristics, not a slice type. The following facts come from [S1], TS 23.501 Table 5.7.4-1 and Notes 4/5/13, using the pinned Release 17 extraction. Later releases exist. Application mappings are project policy: remote surgery uses a discrete-automation proxy; high-voltage smart-grid protection is delay-critical, not mMTC. The mMTC example deliberately chooses non-GBR 5QI 9 for delay-tolerant buffered traffic; Release 17 has no dedicated massive-IoT 5QI in this table.

Profile	5QI	PDB ms	CN PDB ms	PER	CN note
urllc	82	10	1	0.0001	4
v2x	86	5	2	0.0001	5
smart grid	85	5	2	1e-05	5
embb	2	150	20	0.001	13
mmtc	9	300	20	1e-06	13

The CN PDB covers the UPF terminating N6 to 5G-AN segment, closest to the modelled transport network. Method A uses that allocation as the default transport target. Method B, shown only for comparison, is (PDB - CN PDB) times transport_share. The configurable share defaults to 0.5: ASSUMPTION, not standardized. Our review of [S4] and [S5] found interface-dependent latency guidance and slice objectives, not a universal fraction. Giving this emulator the entire CN budget is also an approximation because CN processing can consume part of it.

Profile	A ms default	B ms comparison	Loss % proxy	Minimum Mbps
urllc	1	4.5	0.01	20
v2x	2	1.5	0.01	10
smart grid	2	1.5	0.001	1
embb	20	65	0.1	50
mmtc	20	140	0.0001	None (non-GBR)

Loss percentage is PER times 100, a numerical conversion used as a project IP-loss proxy. PER is a non-congestion link-layer error measure, not IP loss. Late-packet handling also differs by resource type. The broadband rate uses the urban-macro experienced downlink target in [S2] Table 7.1-1; its Note 5 says targets are not strict requirements. Other positive rates are engineering assumptions, not surgical, V2X or protection standards. Non-GBR mMTC requests no minimum rate. Standard priorities are not copied to the project's separate priority scale.

The next table is calculated from the unchanged topology and default method A without existing reservations; it is not measured compliance. No route satisfies any selected delay-critical profile. Their CN reference allocations are smaller than this emulator's per-link delays, so the system must refuse them instead of weakening the targets. For method B, mMTC fits all routes; eMBB still excludes the southern bandwidth bottleneck.

Profile	North A	Middle A	South A
urllc	No	No	No
v2x	No	No	No
smart grid	No	No	No
embb	Yes	Yes	No
mmtc	Yes	Yes	No

Functional checks under docs/demo_assets/profile_* record explained refusals for the delay-critical defaults and actual installed routes plus matching packet delivery for feasible profiles. These checks do not establish sustained bandwidth, one-way latency, or extremely low loss compliance. The live probe still measures RTT whereas planning and the standard use one-way budgets.

Published E1-E8 results used illustrative intents sized for emulation. They remain evaluations of enforcement and recovery mechanisms; they were not rerun or relabelled as standards-compliant services. The original scenario/demo files now carry an explicit illustrative-values comment.

[S1] ETSI TS 123 501 V17.9.0 (2023-07), 3GPP TS 23.501 Release 17, clauses 5.7.3.4/5.7.3.5 and Table 5.7.4-1, Notes 4/5/13.
https://www.etsi.org/deliver/etsi_ts/123500_123599/123501/17.09.00_60/ts_123501v170900p.pdf

[S2] ETSI TS 122 261 V17.10.0 (2022-05), 3GPP TS 22.261 Release 17, Tables 7.1-1 and 7.2.3.2-1. The latter is an ITS infrastructure-backhaul example; its reliability/availability figures are not PER or collision-avoidance targets.
https://www.etsi.org/deliver/etsi_ts/122200_122299/122261/17.10.00_60/ts_122261v171000p.pdf

[S3] ITU-R M.2410-0 (11/2017), clauses 4.7.1, 4.8 and 4.10. Its latency/reliability/density requirements concern radio-interface evaluation; they are context only, not transport inputs.
https://www.itu.int/dms_pub/itu-r/opb/rep/R-REP-M.2410-2017-PDF-E.pdf

[S4] ITU-T GSTR-TN5G (2018-10), section 7.2 / Table 7-2. https://www.itu.int/dms_pub/itu-t/opb/tut/T-TUT-HOME-2018-2-PDF-E.pdf

[S5] RFC 9543 (2024), section 5.1.1.1. https://datatracker.ietf.org/doc/html/rfc9543#section-5.1.1.1

3.2 Parameter provenance and realism

The source-located register docs/PARAMETER_PROVENANCE.md records 566 configured parameter occurrences, including every ordinary scenario/demo intent field, service characteristics, runtime defaults and evaluation workloads. Counts are occurrences, not unique numbers. Papers motivate mechanisms but do not prescribe these numerical constants; configured settings are not measured field data. No experiments or parameters were changed for this audit.

Source type	Rows	Basis	Qualification
STANDARD	36	Pinned QoS catalogue	Mapping remains project policy
LITERATURE	0	Concepts only	No numerical settings attributed
MEASURED	0	No configured field data	Measured results remain separate
PROJECT PLAN	13	Specified link budgets	Not real carrier measurements
ENGINEERING CHOICE	509	Runtime and fixtures	No optimal tuning claimed
ASSUMPTION	8	Rates, share, inferred links	Explicit and revisable

The custom UDP echo times send-to-reply at the source (runtime.py); probes.py and collector.py retain round-trip delay. RFC 7679 section 3.4 defines one-way delay; RFC 2681 section 2.4 defines round-trip delay. RFC 5357 sections 3-4 specify TWAMP control/test exchanges; our probe is not TWAMP. On symmetric paths only, RTT is approximately twice one-way delay plus endpoint overhead. Comparing RTT to the same field used for one-way planning effectively halves the permissible one-way budget. For default profile targets 1/2/2/20/20 ms this implies approximately 0.5/1/1/10/10 ms. The three delay-critical profiles remain infeasible. eMBB/mMTC middle-path planning at 15 ms can be feasible while its nominal 30 ms RTT breaches the unchanged 20 ms detector bound. Profile activation tests therefore establish forwarding, not delay compliance.

For E1-E5's 35 ms latency field, a symmetric interpretation is 17.5 ms one-way. B3's 10% band enters above 38.5 ms RTT and recovers below 31.5 ms RTT, after smoothing and confirmation. E6's 22 ms bound has entry/recovery 24.2/19.8 ms RTT, equivalent to 12.1/9.9 ms one-way under symmetry. E1/E2 describe operational RTT-predicate duration/detection, not normalized one-way compliance. E3 control timings, E4 packet counts (synthetically triggered), E5 rule counts and E6 event accounting remain the recorded mechanism results. No numerical findings were recomputed or relabelled as standards compliance.

ITU-T G.114 (05/2003), Annex A.1 Table A.1, gives a provisional optical-fibre planning value of 5 microseconds/km (Note 1). If entirely propagation, our 2/5/12 ms per-link settings imply 400/1000/2400 km; whole routes imply 1200/3000/4800 km. Netem may instead represent queuing, processing or deliberate stress. These are not measured metro spans. An assumed 10-100 km aggregation comparison would yield 0.05-0.5 ms propagation; an authoritative universal typical-span range was not verified. The topology is deliberately unsuitable for the selected 1-2 ms delay-critical transport targets.

ITU-T Y.1541 (12/2011), clause 5.3 Tables 1-2, provides a closest-use comparator, not a normative 5QI mapping: real-time UDP uses class 0 for comparison (mean one-way IPTD 100 ms, IPDV 50 ms, IPLR 0.1%, IPER 0.01%); TCP transaction profiles use class 2 (100 ms, no IPDV objective, same loss/error bounds). Provisional class 6 (clause 5.3.7 Table 3) has 100 ms, 50 ms, IPLR 0.001%, IPER 0.0001%. Our URLLC/V2X delay/loss targets are numerically stricter than class 0; eMBB equals its loss bound; smart-grid equals provisional class 6 loss with stricter delay; mMTC is stricter than class 2 loss/delay. The register compares all scenario/demo/fixture targets individually. Some illustrative loss bounds, including S1's 0.5%,

are looser than class 0. Our consecutive-RTT jitter is not Y.1541 quantile-minus-minimum IPDV; IPER is not assessed. The scope, statistics and observation intervals differ, so none establishes Y.1541 conformity.

Detector alpha 0.3, 10% hysteresis, three confirmations and B3's 10 s dwell are engineering settings; B2 uses alpha 1, no band, one confirmation and no dwell. Their numeric optimality is unverified. The previously documented fixed 250 ms drain is not current code: controller.py computes max(100 ms, twice old planned delay plus 100 ms), nominally 112/130/148 ms. Reitblatt et al. [5], section 5 (PDF pp.7-8), motivate versioning and draining, not this number. It is not a proven WAN packet-lifetime bound. E1's 15 s is an analysis censor, not a timed removal of the physical fault; E5's executable poll target is 40 ms despite a stale 20 ms comment. S6 overrides middle egresses to 2.5 ms and deliberately uses fast 0.2 ms resets. The full register gives exact source lines, fault replacement magnitudes, schedules, standards URLs, source hashes and honest unverified items.

4 Implementation

M1 parses the request, applies JSON Schema and semantic checks, stores an IntentRecord, and returns a decision with field-specific explanations. A schema rejection is distinct from a valid request rejected by admission.

M2 constructs a residual-capacity graph using persisted reservations and checks feasibility. Priority preemption is accompanied by durable state and removal of the victim's flow rules. M3 resolves endpoint addresses to access switches, enumerates bounded Yen candidate paths, prunes constraints, and selects deterministically. Modelled route bandwidth is the narrowest link and modelled delay is the sum of edge delays.

M4 generates bidirectional OpenFlow matches and actions with per-intent cookies. IPv4 protocol matches include the required Ethernet type. HTB queues use bits per second, derive egress ceilings from topology, and are selected by SET_QUEUE forwarding actions. Strict mode uses the established versioned update procedure; relaxed mode accepts a transient interruption during direct replacement.

M5 runs a separate live process with UDP echo observations and OVS flow-counter deltas. The live probe measures round trip, whereas planning uses configured one-way delays. The detector combines EWMA, a hysteresis band, consecutive confirmation, and a dwell timer. Confirmed events are persisted before Ryu dispatches them. ControlLoop computes an alternative excluding current route edges, installs it, persists the new path, and records a causal explanation.

M6 serves a read-only dashboard and JSON API. It displays intent state, computed or installed routes, recent measurements, and audit entries. The startup seed job provides application examples and labelled simulated samples when samples are absent. Subsequent live samples are displayed separately. Simulated samples never count as evaluation observations. CODE_WALKTHROUGH.md supplies file/function references and the full trace.

5 Evaluation methodology

The baseline contract comes from Section 11 of the supplied project plan. B0 uses static shortest-path forwarding without intents or monitoring. B1 installs intents open loop. B2 uses naive closed-loop recovery with relaxed updates and an immediate detector. B3 uses the full system. A matched-detector relaxed ablation helps distinguish detector cost from update cost. Historical and current-code run families remain separate in RESULT_PROVENANCE.md.

E1 measures violation duration; E2 measures detection latency; E3 measures event-to-install recovery; E4 counts packet loss and reordering around a route change; E5 measures peak rule populations; E6 counts reroutes over a fixed oscillation window; E7 times admission versus active-intent count; E8 measures control traffic and CPU. O3 separately checks data-plane queue binding under competing traffic. API response latency is not switch-install latency. E8's capture-based traffic field estimates control traffic using TCP PSH segments; it is not a parser that proves each segment contains exactly one OpenFlow message.

Each registered headline family uses independent repetitions, with its n shown in the generated tables. Intervals are two-sided Student-t intervals using sample standard deviation. Welch comparisons are used for nonzero variances; exact two-sided permutation comparisons are used when a group has zero sample variance. Deterministic repeated observations do not justify an infinite t statistic with a reported probability of zero. Open-loop violation duration is explicitly right-censored at the observation boundary.

Experiments run inside the Linux Docker image under WSL on the Windows host, using the project's pinned Python/Ryu stack and OVS software datapath. No claim is made about physical switching hardware. Raw CSV files are retained; changed-code runs and regression checks use distinct prefixes. An independent verifier checks registered summaries against raw values. The E6 event-accounting follow-up is documented with direct SQL counts and complete-sample checks; historical uncertain event counts are not treated as established causal evidence.

6 Results

The following tables are generated from registered summary files. They report measured means, sample deviations, and intervals rather than values transcribed into the builder. Historical baselines remain identified where current-code reruns are compared with them. The corrected E6 durable accounting series (n=10) is included alongside the retained oscillation series, verifying that each reroute matches an emitted durable violation event.

B0 static forwarding baseline

Source: experiments/results/summary/b0_baseline_summary_20261025.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B0	loss	10	0.000	0.000	[0.000, 0.000]	%
B0	rtt avg	10	12.271	0.046	[12.238, 12.304]	ms
B0	throughput	10	71.590	0.032	[71.567, 71.613]	Mbits/s

B1 API response timing pilot

Source: experiments/results/summary/b1_open_loop_20261025.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B1	intent post response latency	10	61.982	6.613	[57.251, 66.712]	ms

E1 violation duration

Source: experiments/results/summary/e1_violation_duration_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B1	sla violation duration s	10	15.000	0.000	[15.000, 15.000]	s
B3	sla violation duration s	10	3.003	0.000	[3.003, 3.003]	s

Historical E2 and E3 baselines

Source: experiments/results/summary/b2_b3_closed_loop_20261025.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B2	detection latency	10	99.467	4.498	[96.249, 102.684]	ms
B2	recovery latency	10	279.355	70.552	[228.889, 329.821]	ms
B2 matched detector	detection latency	10	2116.845	13.402	[2107.259, 2126.432]	ms
B2 matched detector	recovery latency	10	467.711	59.878	[424.880, 510.542]	ms
B3	detection latency	10	2104.262	3.994	[2101.405, 2107.119]	ms

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B3	recovery latency	10	618.535	82.519	[559.508, 677.561]	ms

Current code B3 E2 and E3

Source: experiments/results/summary/b3_current_closed_loop_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B3-current	detection latency	10	2115.940	10.073	[2108.734, 2123.145]	ms
B3-current	recovery latency	10	537.853	127.828	[446.416, 629.289]	ms

Current code E4 transition behavior

Source: experiments/results/summary/e4_current_packet_loss_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B2	packets lost	10	41.000	2.160	[39.455, 42.545]	packets
B2	packets reordered	10	0.000	0.000	[0.000, 0.000]	packets
B3	packets lost	10	0.000	0.000	[0.000, 0.000]	packets
B3	packets reordered	10	28.000	0.000	[28.000, 28.000]	packets

Historical E5 rule and loss tradeoff

Source: experiments/results/summary/e5_flow_table_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
all relaxed	packets lost	10	56.800	2.394	[55.087, 58.513]	packets
all relaxed	peak rules max switch	10	6.000	0.000	[6.000, 6.000]	entries
all strict	packets lost	10	0.000	0.000	[0.000, 0.000]	packets
all strict	peak rules max switch	10	10.000	0.000	[10.000, 10.000]	entries
selective	packets lost	10	33.900	1.792	[32.618, 35.182]	packets
selective	peak rules max switch	10	9.000	0.000	[9.000, 9.000]	entries

E6 retained oscillation series

Source: experiments/results/summary/e6_oscillation_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B2	dwell suppressions	10	0.000	0.000	[0.000, 0.000]	count
B2	reroute count 10min	10	90.000	0.000	[90.000, 90.000]	per 10 min
B3	dwell suppressions	10	28.700	2.111	[27.190, 30.210]	count
B3	reroute count 10min	10	48.300	1.059	[47.542, 49.058]	per 10 min

E7 admission scaling

Source: experiments/results/summary/e7_admission_scaling_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
k=01	admission decision latency ms	10	47.889	3.789	[45.179, 50.599]	ms
k=02	admission decision latency ms	10	49.420	4.427	[46.253, 52.587]	ms
k=04	admission decision latency ms	10	49.665	2.123	[48.146, 51.184]	ms
k=08	admission decision latency ms	10	51.845	2.109	[50.336, 53.354]	ms
k=16	admission decision latency ms	10	59.496	8.463	[53.442, 65.550]	ms

E8 corrected control stack overhead

Source: experiments/results/summary/e8v2_control_overhead_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B1	control stack cpu pct	10	0.939	0.045	[0.907, 0.971]	%
B1	openflow msgs per s	10	8.194	0.073	[8.142, 8.246]	msg/s
B3	control stack cpu pct	10	5.078	0.173	[4.954, 5.201]	%
B3	openflow msgs per s	10	9.323	0.117	[9.240, 9.407]	msg/s

Current code O3 bound queue

Source: experiments/results/summary/o3_current_queue_binding_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
bound	intent loss pct	10	0.000	0.000	[0.000, 0.000]	%
bound	intent throughput mbps	10	24.995	0.000	[24.995, 24.996]	Mbps

Historical O3 bound and unbound comparison

Source: experiments/results/summary/o3_queue_binding_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
bound	intent loss pct	10	0.000	0.000	[0.000, 0.000]	%
bound	intent throughput mbps	10	24.995	0.000	[24.995, 24.996]	Mbps
unbound	intent loss pct	10	23.620	1.529	[22.527, 24.714]	%
unbound	intent throughput mbps	10	19.091	0.382	[18.818, 19.365]	Mbps

Corrected E6 durable accounting

Source: experiments/results/summary/e6_accounting_summary.csv

Group	Metric	n	Mean	Sample SD	95% t interval	Unit
B2	duration s	10	600.000	0.000	[600.000, 600.000]	s
B2	measurement count	10	600.000	0.000	[600.000, 600.000]	count
B2	raw reroute count	10	90.000	0.000	[90.000, 90.000]	count
B2	reroute count 10min	10	90.000	0.000	[90.000, 90.000]	per 10 min
B2	violations emitted	10	90.000	0.000	[90.000, 90.000]	count

E4 exact comparisons

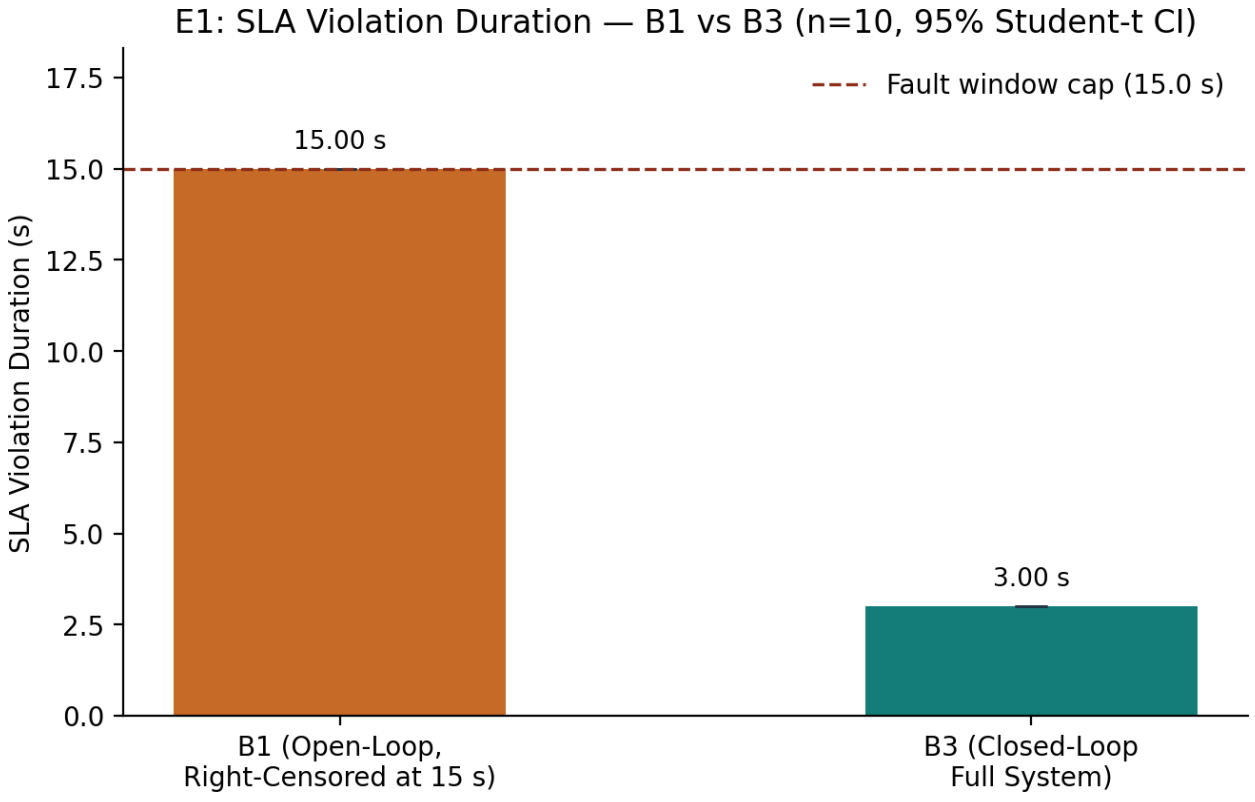
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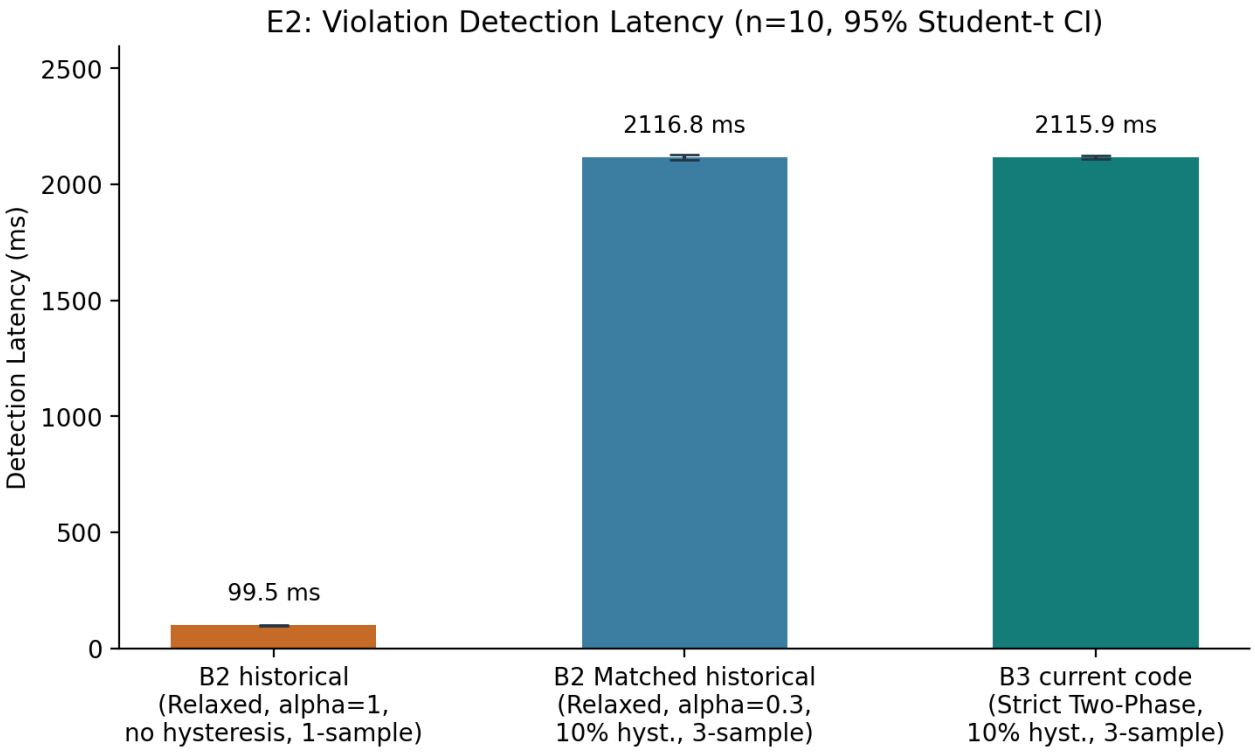
Captured runtime environment

Python 3.9.25; Ryu 4.34; Mininet 2.3.1b4; ovs-vsctl (Open vSwitch) 2.17.12; kernel 6.18.33.1-microsoft-standard-WSL2. Captured by docs/capture_report_environment.py at 2026-11-02T19:46:12.859687+00:00.

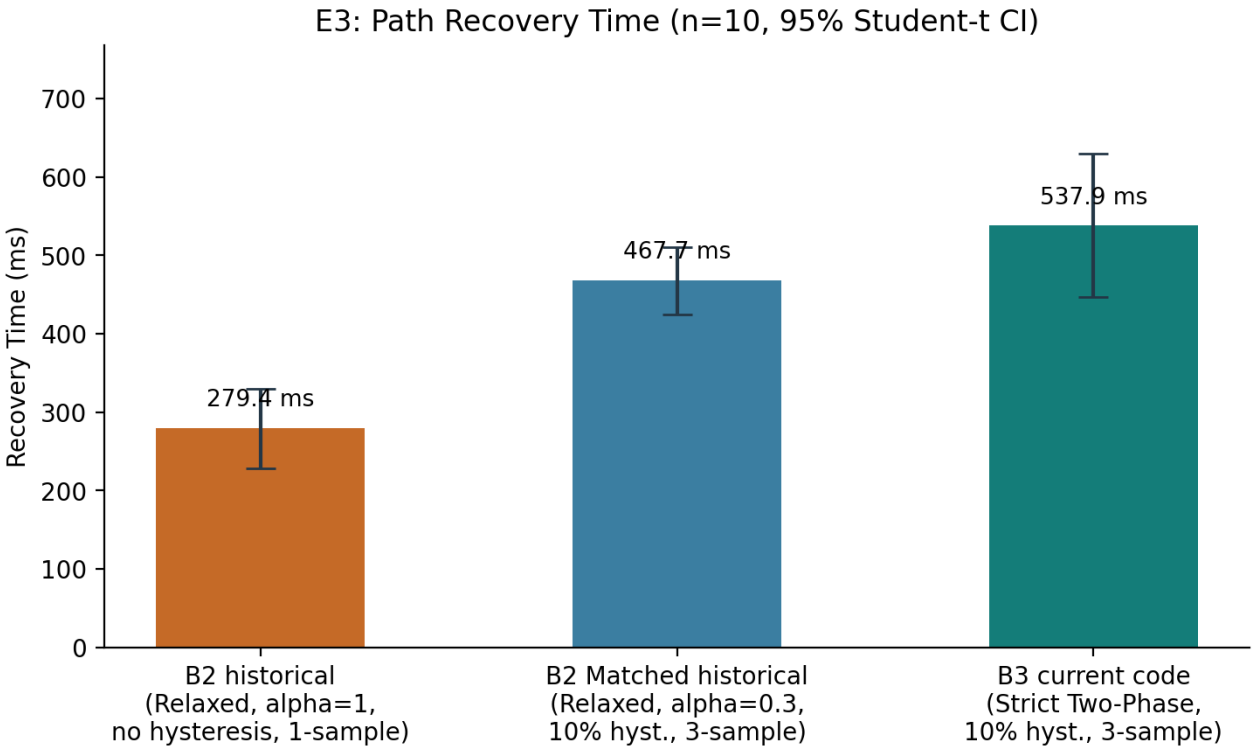
Evaluation figures



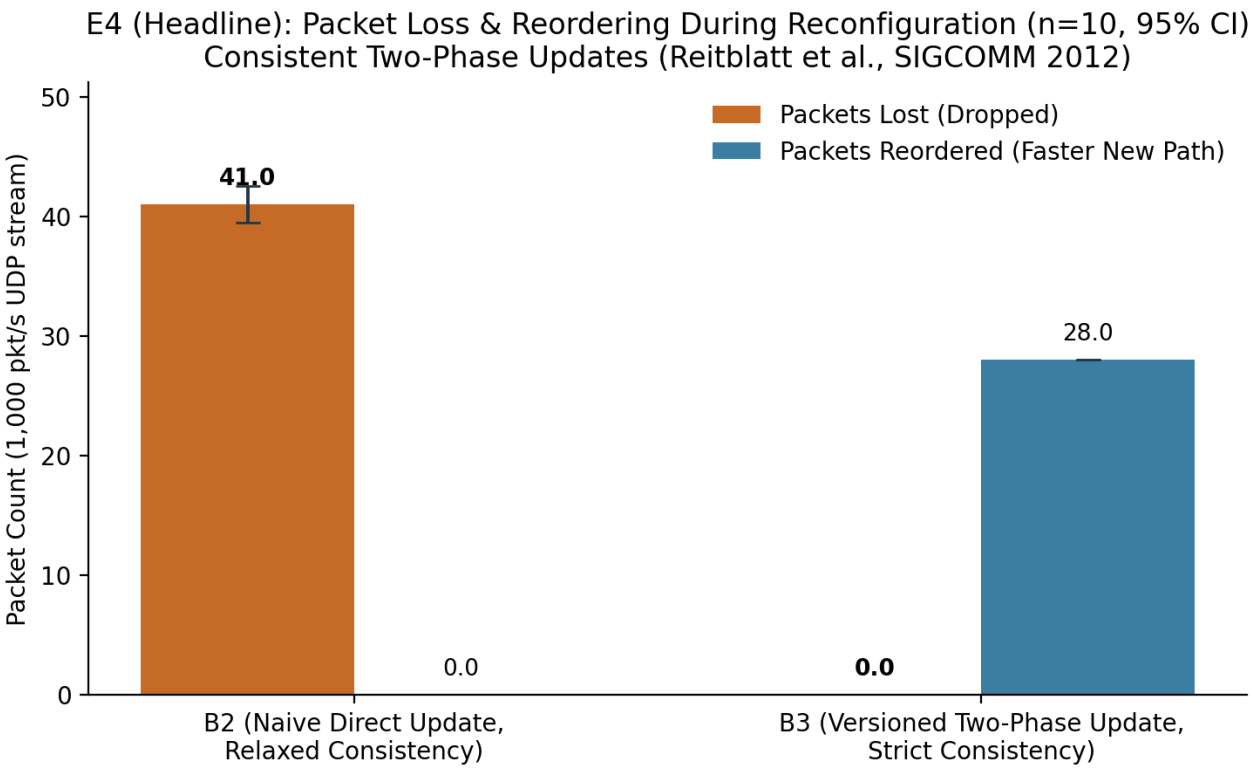
Violation duration with open-loop censoring



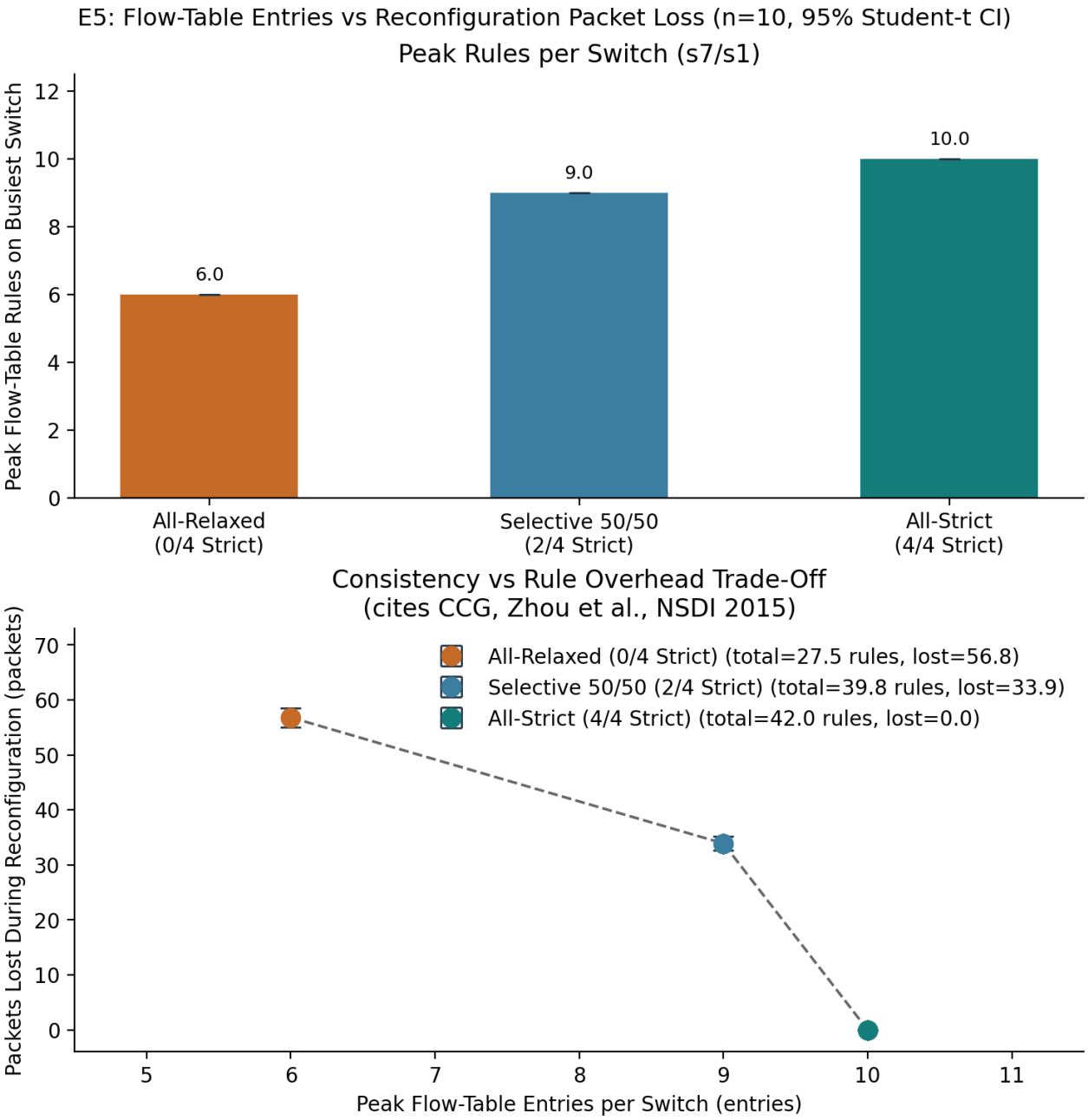
Detection latency with code-version provenance



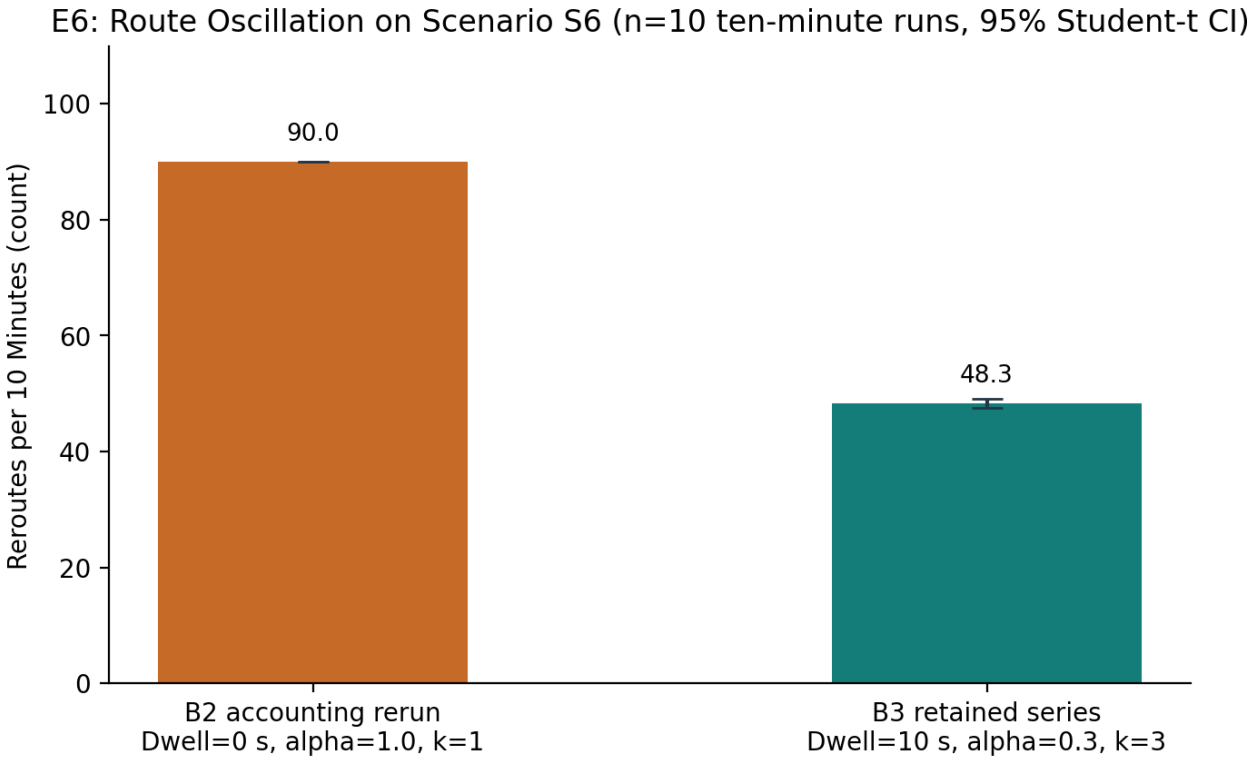
Recovery latency with code-version provenance



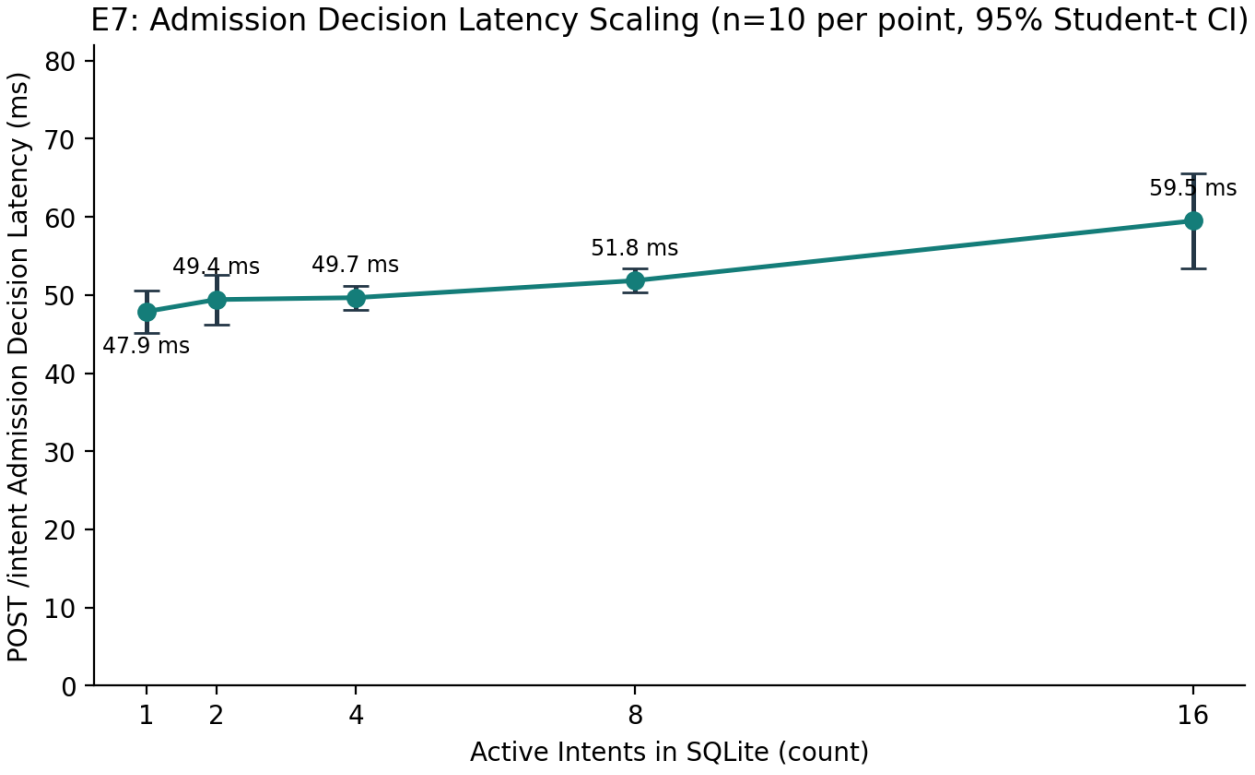
Transition packet loss and reordering



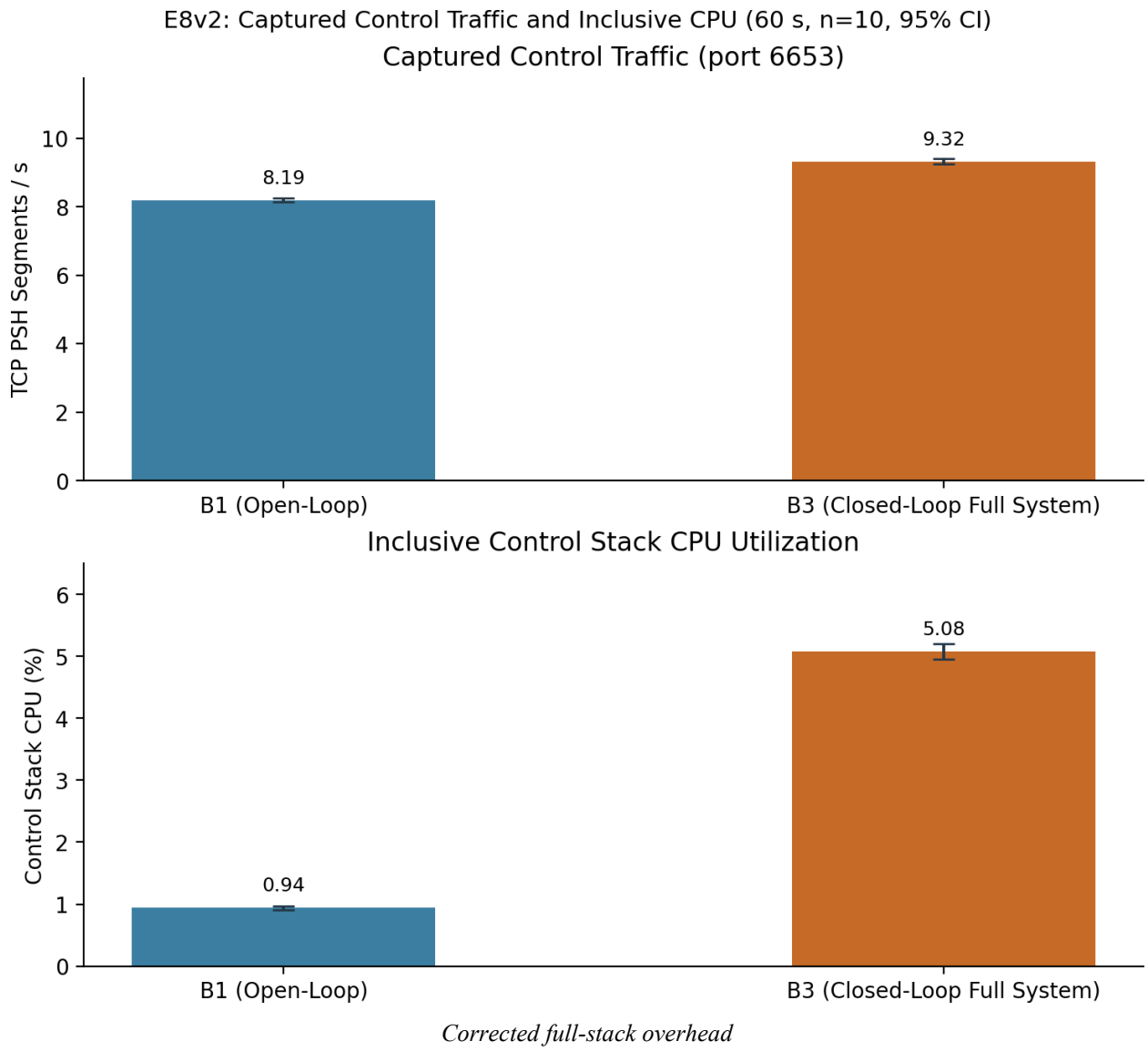
Historical E5 rule-space and packet-preservation tradeoff; mixed code versions



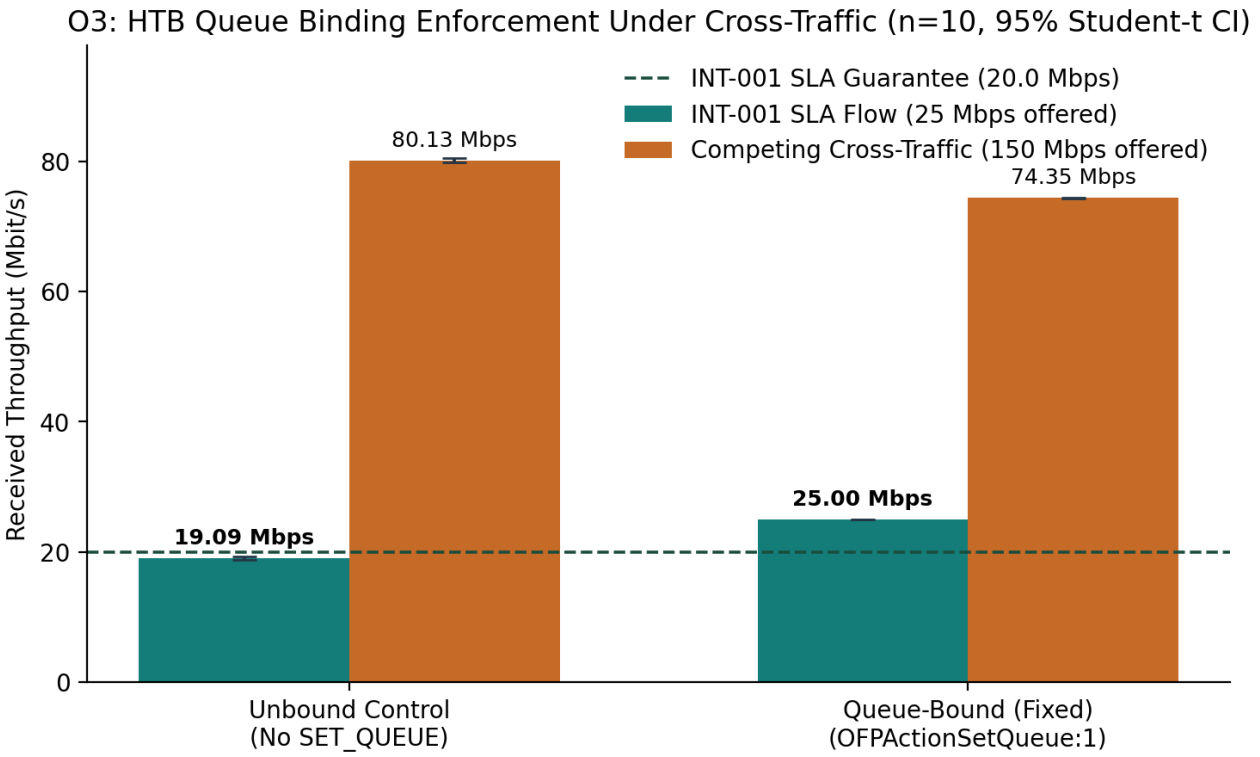
Oscillation over the fixed observation window; series provenance is labelled



Admission decision scaling



Corrected full-stack overhead



Historical O3 queue-binding comparison under competing traffic

7 Discussion

Strict updates preserve packets in the tested transitions, while damping reduces route changes during alternating faults. Those benefits have measurable costs: confirmation delays detection, staging and draining delay installation, new packets can overtake old packets, and overlapping versions consume extra rules. The corrected mean B3-to-B1 control-stack CPU ratio is 5.408, calculated from E8v2's measured group means. It must not be interpreted as whole-machine CPU or extrapolated to a production deployment.

Low-rate probes establish reachability and detector behavior but do not load-test a bandwidth promise. O3 therefore offers sustained protected traffic with competing load and measures receiver throughput. The E4 interpretation is consistent with the configured path-delay difference and sequence rate; this is a timing explanation rather than a new theoretical guarantee. For E6, the historical event-count shortfall is addressed by the corrected durable accounting rerun (n=10, 90 reroutes reconciling exactly with 90 durable violations and 90 restorations per run; see E6_ACCOUNTING.md). Historical runs remain disclosed as a provenance limitation, while the oscillation comparison remains a path-revision comparison.

8 Limitations

The fixed topology and single-machine emulation limit generalization. Inferred links are design assumptions. Candidate search is bounded. Planning checks a one-way link-delay sum while live detection compares UDP round-trip delay with the same latency field; the implementation does not convert that budget between the two definitions. Its measurements therefore do not certify a common one-way latency SLA. A software-switch table pilot does not demonstrate hardware TCAM pressure. The drain interval is calibrated for the experiment, not a proven WAN lifetime bound. Forwarding selects a shared queue id on an egress port, so the single-protected-flow bandwidth experiment does not establish independent per-intent bandwidth isolation for concurrent services. Recovery computes an alternative directly on its topology graph without repeating M2 residual admission; contention-safe multi-intent recovery is not established. Queue reservations do not alone prove sustained delivered application throughput. Exact tests and confidence intervals describe these runs and do not remove workload, platform, or code-version confounders. Simulated dashboard data is for presentation only.

9 Conclusion and future work

The project implements a reproducible intent-to-enforcement and measurement loop with explainable admission, durable violation handling, and per-intent update consistency. Its evaluated strengths and costs are both visible in the generated results. The contribution is an integrated engineering artifact based on cited methods. Future work includes larger and changing topologies, controlled hardware experiments, stronger workload validation, durable provenance for each run, and validation of update-drain assumptions under wider delay distributions.

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